

What is already known/ what does this study add?

Despite decades of public health interventions, smoking continues to contribute substantially to the global burden of disease, including cardiovascular diseases, chronic respiratory conditions, and various malignancies. Various smoking cessation strategies exist including Nicotine Replacement Therapy (NRT), pharmacological therapies such as varenicline and bupropion, behavioral counseling, and unassisted quitting.

The aim of this study is to assess the **effectiveness** of Nicotine Replacement Therapy and Other Smoking Cessation Methods Compared with No Intervention in Achieving successful Smoking Cessation: A **Case-Control** Study.

Introduction **Redundant - three paragraphs - 800 words max**

Tobacco smoking remains one of the most significant preventable causes of morbidity and mortality worldwide, According to WHO, each year, tobacco causes 7 million deaths. Current trends show that by 2030, tobacco is likely to be the world's leading cause of death and disability, killing more than 10 million people annually.

In Egypt, 22.8% of Egyptians are currently tobacco smokers, 18.4% are daily tobacco smokers, the commonest form of tobacco use is cigarettes, with average age starting smoking 18.2. [1]

Cigarettes contain 4000 hazardous substances, The most important and active psychopharmaceutical drug contained in the leaves of the tobacco plant is nicotine.

Nicotine dependence is a complex chronic condition characterized by behavioral, psychological and physical components. The addictive potential of nicotine is primarily mediated through stimulation of $\alpha 4\beta 2$ nicotinic acetylcholine receptors in the brain, resulting in the release of neurotransmitters such as dopamine within the mesolimbic reward pathway.

This neurobiological mechanism reinforces continued tobacco use and contributes to withdrawal symptoms when smoking is discontinued. These withdrawal symptoms reach its peak in 2-3 days and may include irritability, anxiety, difficulty concentrating, depressed mood, and strong cravings for nicotine, which collectively contribute to high relapse rates among smokers attempting to quit. [3]

Smoking should be regarded as a primary addictive disorder, this contrasts with the classic view of smoking as a habit or life style choice, it has been estimated that 75% of American smokers wish to quit but only 3% are able to achieve prolonged abstinence in any year, which indicates the involuntary nature of the established addiction. In addition, current concepts suggest smoking should be regarded as a chronic relapsing disorder. In this context, a cessation attempt should be regarded as an attempt to induce a remission. The abstinent smoker, moreover, should always be viewed as at risk for relapse. The goal for therapy is to induce a remission that is as durable as possible. [2]

Smoking cessation is a cost-effective treatment, about £2000 per quality-adjusted life year (QALY) for patients with chronic obstructive pulmonary disease (COPD). According to NHS, the UK government has set targets to reduce the numbers of smokers, with substantial funding for smoking cessation services about £66.4 million in 2011.

Over the past several decades, numerous interventions have been developed to assist individuals in quitting smoking, these approaches include nicotine replacement therapy (NRT), non-nicotine pharmacotherapies such as bupropion and varenicline, behavioral counseling. Each method targets different components of nicotine dependence and varies in terms of effectiveness, accessibility, and patient acceptability. To achieve sustained abstinence the aims are to reduce short-term nicotine cravings and to modify behavior in the long term.

Nicotine replacement therapy (NRT) is one of the most widely used and well-established pharmacological approaches to smoking cessation. The concept is to replace nicotine that would be absorbed from cigarettes and thereby reduce the intensity of withdrawal. However, Smokers will experience withdrawal symptoms with less intensity. NRT is available in several formulations, including transdermal patches, chewing gum, lozenges, inhalers, and nasal sprays. Evidence from randomized controlled trials and meta-analyses indicates that NRT increases smoking cessation rates by approximately 1.5 to 2 times compared with placebo or no treatment, and one trial comparing gum, inhaler, and nasal spray found no difference in efficacy. However, they differ in their pharmacokinetics. Transdermal systems provide the slowest delivery of nicotine but maintain steady-state levels throughout the day. The other formulations allow episodic dosing. A common practice is to combine a transdermal system with another formulation, a “patch-plus” regimen. This allows a

smoker to increase nicotine delivery at times of urges. Clinical trial data support better success with combined NRT compared to monotherapy. [4]

In addition to nicotine replacement, several non-nicotine pharmacological agents have been developed to support smoking cessation. Among these, Bupropion and Varenicline are widely recognized as effective pharmacotherapies. Bupropion, originally developed as an antidepressant, acts by inhibiting the reuptake of dopamine and norepinephrine in the brain. This mechanism reduces nicotine cravings and withdrawal symptoms while also attenuating the reinforcing effects of nicotine. Clinical trials have demonstrated that bupropion significantly increases smoking abstinence rates compared with placebo and shows effectiveness comparable to single-form NRT when combined with behavioral support. The most common adverse effects are dry mouth, insomnia, agitation, and headache. [5]

Varenicline represents another important pharmacological intervention for smoking cessation. It acts as a partial agonist at the $\alpha 4\beta 2$ nicotinic acetylcholine receptor, simultaneously stimulating the receptor to reduce withdrawal symptoms while blocking nicotine from binding and producing its reinforcing effects. Evidence from randomized controlled trials indicates that varenicline is more effective than placebo, bupropion and single-form NRT in achieving sustained smoking abstinence. Consequently, many clinical guidelines recommend varenicline as a first-line pharmacological treatment for smokers who wish to quit. The most common adverse reactions are nausea, insomnia, visual disturbances, syncope, and skin reactions. [6]

According to FDA both bupropion and varenicline had a possible association with suicidal events. Thus, it is important for all patients and their caregivers to monitor for possibility of neuropsychiatric symptoms, and patients should be monitored for changes in behavior, hostility, agitation, depressed mood, suicidal ideation, and suicide attempts.

While pharmacotherapy addresses the biological aspects of nicotine dependence, behavioral interventions play a critical role in addressing the psychological and social factors associated with smoking behavior. Behavioral therapy encompasses a wide range of approaches, including individual counseling, group therapy, motivational interviewing, cognitive behavioral therapy, and structured

smoking cessation programs. These interventions aim to help smokers identify triggers for tobacco use, develop coping strategies to manage cravings, and maintain long-term abstinence. Evidence suggests that behavioral interventions significantly increase the likelihood of successful cessation, particularly when combined with pharmacological therapies. Studies have demonstrated that intensive behavioral support can further enhance quit rates among individuals receiving NRT or pharmacotherapy. [2]

Despite the availability of these evidence-based interventions, many smokers attempt to quit without any formal assistance, a strategy often referred to as unassisted or “cold turkey” quitting. Unassisted quitting remains one of the most common cessation methods worldwide due to factors such as limited access to healthcare services, lack of awareness about available treatments, or personal preference for non-pharmacological approaches. Although some individuals successfully achieve long-term abstinence through unassisted attempts, evidence suggests that success rates are generally lower compared with interventions that include pharmacotherapy or behavioral support.

Aim/Objectives

Comparative evaluation of different smoking cessation methods including nicotine replacement therapy, pharmacological agents such as bupropion and varenicline, behavioral therapy, and unassisted quitting remains essential to inform clinical practice and optimize smoking cessation programs. Understanding the relative effectiveness of these approaches can help guide healthcare providers in selecting the most appropriate interventions for smokers and improve overall cessation outcomes.

Methodology

- **Type of study:** Case-Control Study.
- **Study setting:** The study will be carried out at smoking cessation clinic at Ain Shams University Hospitals.

- **Study Population:**
 - Consecutive sampling of patients visiting smoking cessation clinic at Ain Shams University Hospitals according to the mentioned inclusion and exclusion criteria.
 - Cases: people who successfully quit smoking for more than 6 months and are not smokers at the time of the study.
 - Controls: people who failed to quit smoking or relapsed in less than 6 months.
 - Inclusion Criteria:
 - 1- Adult aged 18 years or older.
 - 2- History of regular cigarette smoking (more than 100 cigarettes).
 - 3- Attempted smoking cessation at least once using one of the following methods: Nicotine replacement therapy (NRT), Pharmacological therapy (bupropion or varenicline), Behavioral therapy, Unassisted quitting.
 - 4- Willingness to participate in the study and provide informed consent.
 - Exclusion criteria:
 - 1- Have incomplete medical records or insufficient information about the cessation method used.
 - 2- Used multiple cessation interventions simultaneously that cannot be clearly categorized.
 - 3- Have severe cognitive impairment or psychiatric conditions that interfere with reliable data collection.
 - 4- Non-smokers.
 - 5- Smokers who never attempted smoking cessation before.
- **Study size:** Sample size calculation was performed using EPITOOLS calculator with significance level (Alpha) of 0.05 and Power of 80%.
- The prevalence among unassisted quitting approximately 5%. With expected odds ratios in group 2 (behavioral), 3 (NRT) and 4 (bupropion or varenicline) are: 3.35, 4.75 and 6.3, respectively.
- **Sample size was calculated to be 396 participants.**
- **Exposure:** Participants will be categorized based on the smoking cessation method used: No assistance, Behavioral therapy, Nicotine Replacement Therapy (NRT), Pharmacological therapy (varenicline or bupropion).

- **Outcomes:** self-reported Smoking cessation (abstinence ≥ 6 months).
- **Potential confounders:** age, sex, socioeconomic status, **smoking index**, comorbidities, number of previous quit attempts. Smoking index must be explicitly defined
- **Ethical considerations:** Approval from the institutional ethical committee of faculty of medicine Ain Shams University, informed consent from all participants will be taken, data will be anonymously reviewed and analyzed by only the team members of the project.
- **Study tools:** history taking from all participants about age, sex, smoking index, current smoking status, last time he/she smoked, type of tobacco used, any other substances used at time being, number of previous quit attempts, which method is used at this attempt of quitting, any comorbidities or psychiatric illnesses.

References

1. WHO STEPs survey noncommunicable disease risk factor surveillance, EGYPT. 2017.
2. Fiore MC, Jaén CR, Baker TB, et al. Treating tobacco use and dependence: 2008 update. Rockville (MD): US Department of Health and Human Services; 2008.
3. Gotti C, Clementi F, Fornari A, et al. Structural and functional diversity of native brain neuronal nicotinic receptors. *Biochem Pharmacol.* 2009;78(7):703–711.
4. Tahiri M, Mottillo S, Joseph L, et al. Alternative smoking cessation aids: a meta-analysis of randomized controlled trials. *Am J Med.* 2012;125(6):576–584
5. Hughes JR, Stead LF. Antidepressants for smoking cessation. *Cochrane Database Syst Rev.* 2007;(1):CD000031.
6. Coe JW, Brooks PR, Vetelino MG, et al. Varenicline: an $\alpha 4\beta 2$ nicotinic receptor partial agonist for smoking cessation. *J Med Chem.* 2005;48(10):3474–3477.

Data collection sheet

Duration of abstinence (months)
Verification of abstinence (CO/cotinine if available)

Personal data

Patient's hospital ID: -----

Date of birth (MM/DD/YYYY): -----/-----/-----

Gender: [] Male / [] Female

Education Level: [] Primary / [] Secondary / [] University / [] Postgraduate

Employment Status: [] Employed / [] Unemployed / [] Retired / [] Student

Clinical data

Chronic obstructive pulmonary disease: ☐ Yes / ☐ No

Cardiovascular Disease: ☐ Yes / ☐ No if yes please mention it: -----

Psychiatric Illness: ☐ Yes / ☐ No if yes please mention it: -----

Other comorbidities: -----

Smoking History

Have you ever smoked any form of tobacco: [] Yes / [] No

- If yes, which form of tobacco: [] Cigarettes / [] Waterpipe (Shisha) / [] E-cigarettes / [] Vaping / [] Other, (please mention it:-----)
- for how long you have been smoking (in years): -----
- How many cigarettes you smoked per day: -----

Are you a current smoker: [] Yes / [] No

- If no, what is the last time you smoked: [] ≤ 1 week / [] > 1 week - < 1 month / [] 1 – 6 months / [] > 6 months

Have you ever tried to quit smoking: [] Yes / [] No

- If yes how many times you tried to quit: [] 1 time / [] 2-3 times / [] More than 3 times
- Which method of the following did you use (you can choose more than one):
☐ No assistance / ☐ Behavioral therapy / ☐ Nicotine Replacement Therapy (NRT) / ☐ Pharmacological therapy (Varenicline / Bupropion) / ☐ other, (please mention it: -----)
- For the last three methods, for how long you continued on it (in weeks): -----

Design "case-control" while drawing conclusions about comparative effectiveness.

Sample size:

Base the sample size on published quit rates, not assumed ORs.

Specify the primary comparison (e.g., NRT vs unassisted quitting, or pharmacotherapy vs unassisted quitting).

Provide the reference from which the expected quit rates were derived.

State that the study is powered for the smallest clinically relevant difference, making the calculation transparent and conservative.

Importance of research question: 9/10

Feasibility: 9/10

Methodological rigor: 5/10